

Introduction to Python

1. What is Python?

- Python is a high-level programming language known for its simple, readable syntax.
- Allows for writing clean code with fewer lines.
- Supports multiple programming paradigms: object-oriented, functional, and procedural.
- Dynamically typed (no explicit variable declarations) and includes automatic garbage collection.

2. A Glimpse into Python Programming

2.1 Print().

- The `print()` function is a built-in tool used to display text on the screen.
- Text must be wrapped in single or double quotes.

Example

```
Python
print("Hello, World!")
print('Python is fun!')
```

2.2 Code Organization (Indentation)

- Unlike languages using curly braces (e.g., C++, Java), Python uses indentation (spaces or tabs) to define code blocks.
- The visual structure dictates how the code runs.

Example

Python

```
if True:
    print("This is indented and belongs to the block above.")
print("This is not indented and is outside the block.")
```

2.3 Comments

- Used to annotate code
- The hashtag symbol (#) initiates a comment.
- Comments are ignored by the computer during execution.

Example

Python

```
# This is a comment explaining the code below
print("Hello!")
# Output: Hello
```